

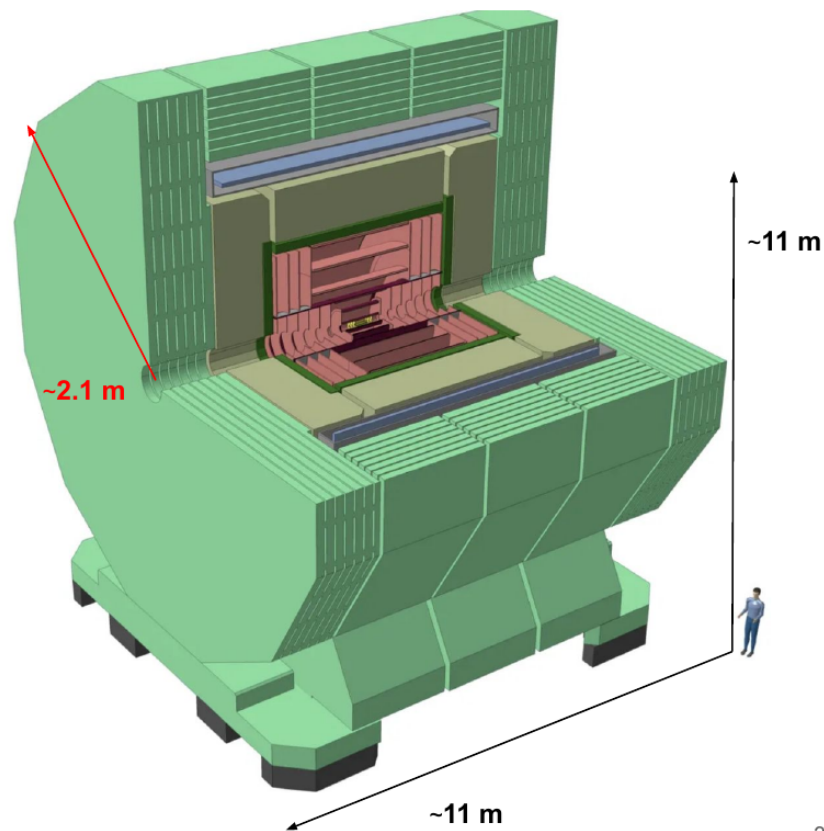
Task A & Task B: SM Identification and BSM Validation

FCC-ee with the CLD Detector

FCC-ee CLD Collaboration

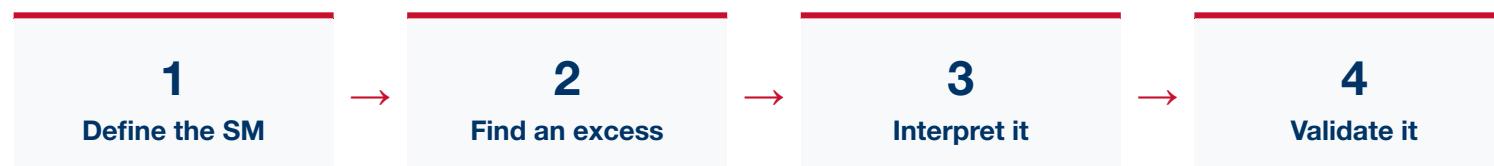
Saurav Bania, Greta Brianti, Jurjan Bootsma, Vincenzo Del Piano, Kobe Degeetere, Andrea Maria, Sergei Solokhin

- FCC-ee: a 100 km e^+e^- collider. Four energies: **91** (Z pole), **160** (WW), **240** (ZH), **365 GeV** ($\bar{t}t$).
- CLD gives us tracking, calorimetry and a 2 T field — enough to reconstruct jets, leptons and missing energy.
- Each energy hands us several **anonymised** samples. We don't know what they are yet.



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CLD detector — original FCC-ee CLD collaboration deck



At **91** and **365 GeV** we found an excess — all four steps apply. At **160** and **240 GeV**, no excess was found. We stop after step 1.

SM process identification at 160 GeV

Reconstructing the known processes before studying the excess

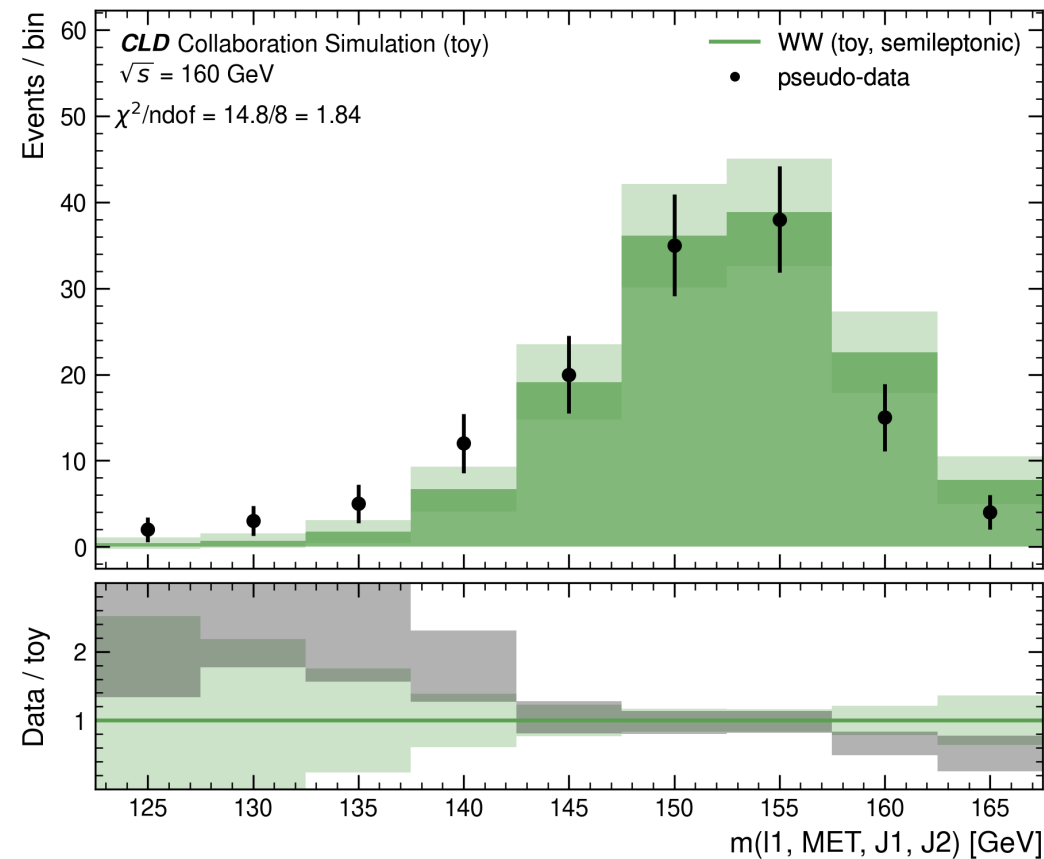
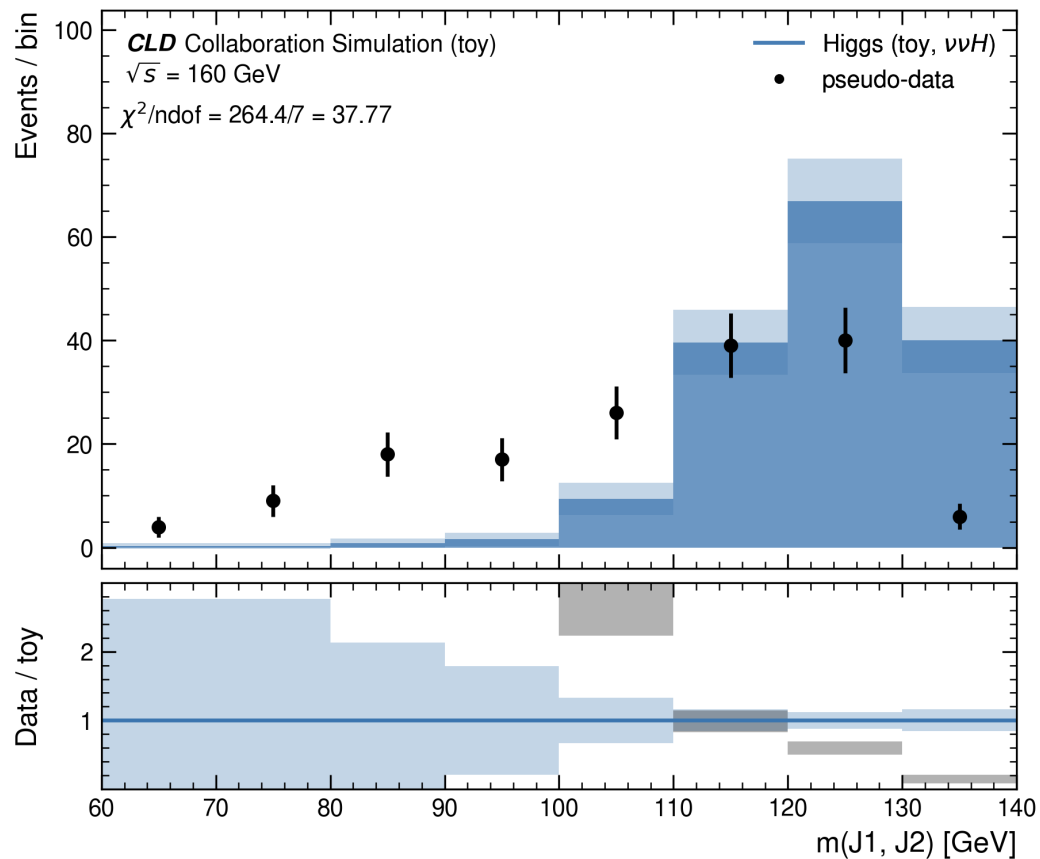
Next: **Kobe Degeetere**

Task A: match each anonymous sample to a known process, using cuts on jets, leptons, b-tagging and MET. Two of four samples are identified this way. No excess is searched for at this energy — step 1 of the method is all we need here.

Process	Key cuts	Efficiency
Higgs (vvH, H→b ̄ b)	2 jets b-tag>0.7, 0 leptons, MET cut	32.7%*
WW (semileptonic)	2 jets, 1 lepton, MET p _T >5 GeV	43.2%*

* assumed (toy), not measured — the talk gives cuts, not absolute yields.

Higgs: the dijet mass peaks at **~125 GeV**. WW: the effective mass recovers close to **$\sqrt{s} = 160$ GeV**, as expected for a semileptonic decay. Both agree with the reported points at the peak.



SM process identification at 365 GeV

Five anonymous samples, five SM assignments

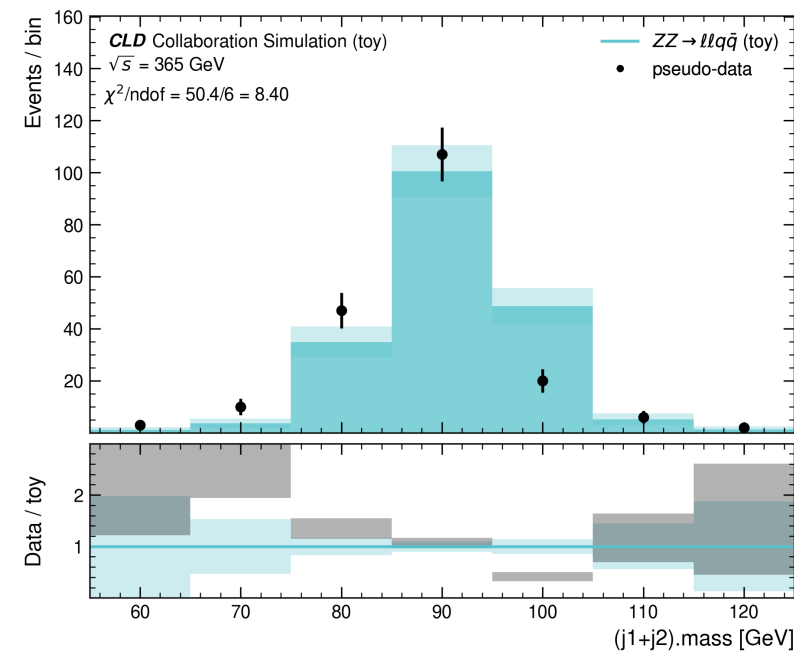
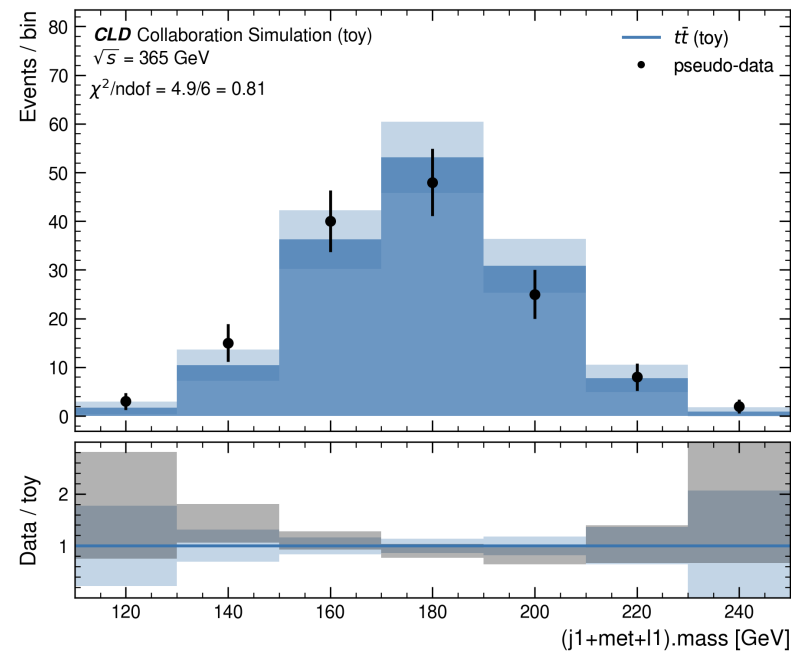
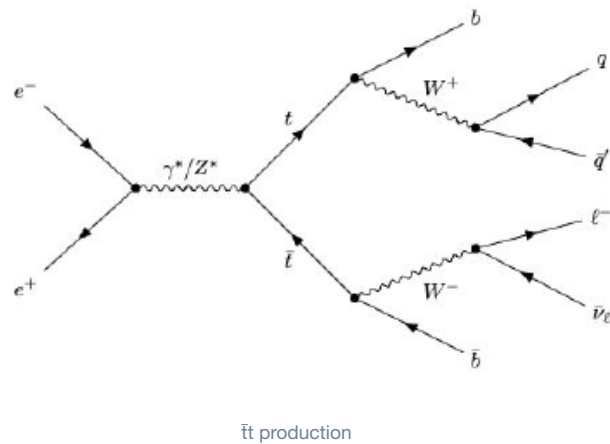
Next: **Saurav Bania**

365 GeV is the most complete working point: all five samples get an assignment — $\bar{t}t$, $e^+e^- \rightarrow \bar{f}f$, ZZ, ZH, WW. More jets are available here than at 160 GeV, so b-tagging separates these final states well.

Process	Efficiency
$\bar{t}t$	10.7%*
$e^+e^- \rightarrow \bar{f}f$	10.2%*
$ZZ \rightarrow \ell\ell q\bar{q}$ (X5)	8.7%*
ZH	5.4%*
WW	52.0%*

* assumed (toy), not measured.

$t\bar{t}$: reconstructed mass sits at the top scale, $\chi^2/\text{ndof} = 0.81$ — the best agreement in this study. ZZ: dijet mass peaks cleanly at $m_Z = 91 \text{ GeV}$.

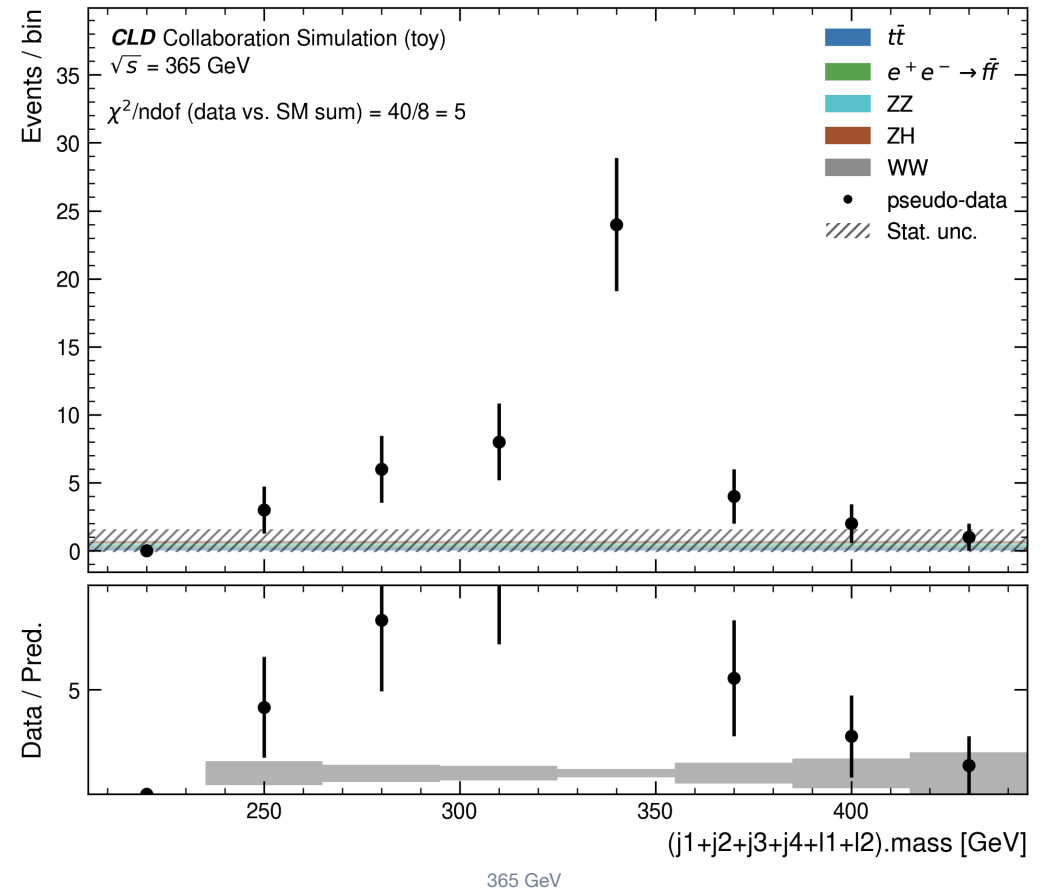
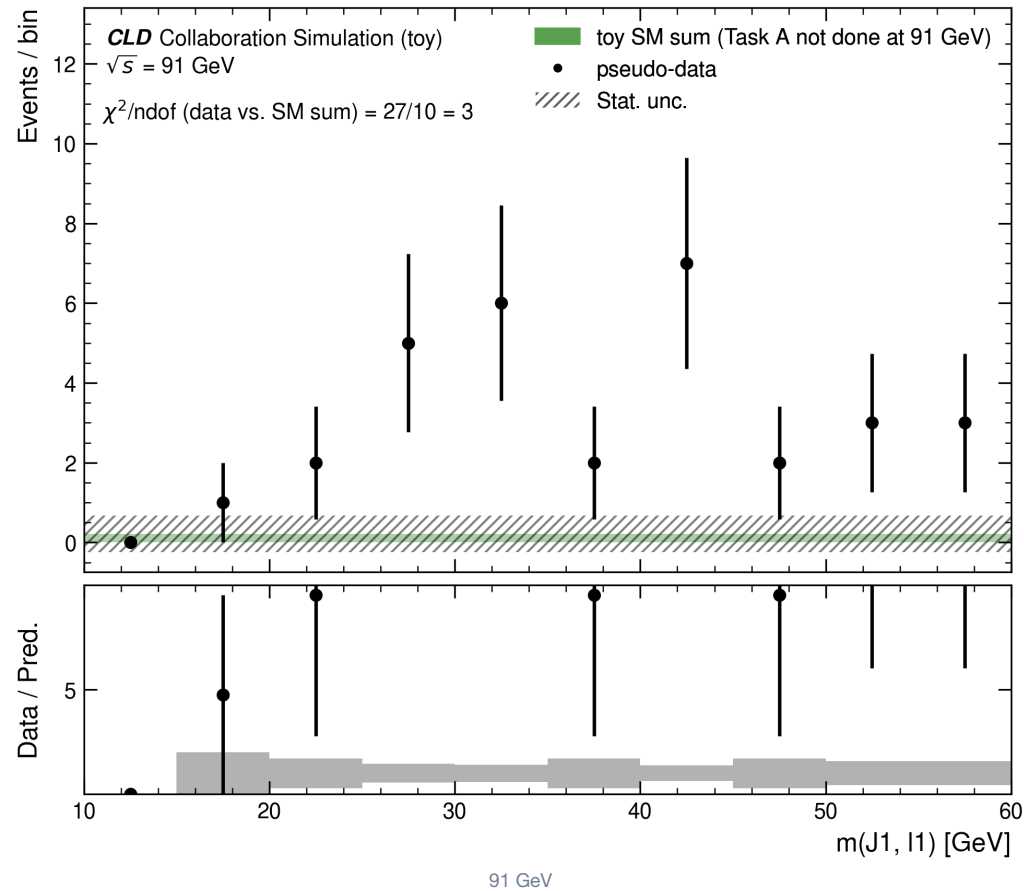


From SM identification to the excess

Once the samples are identified, can the SM explain the observed structure?

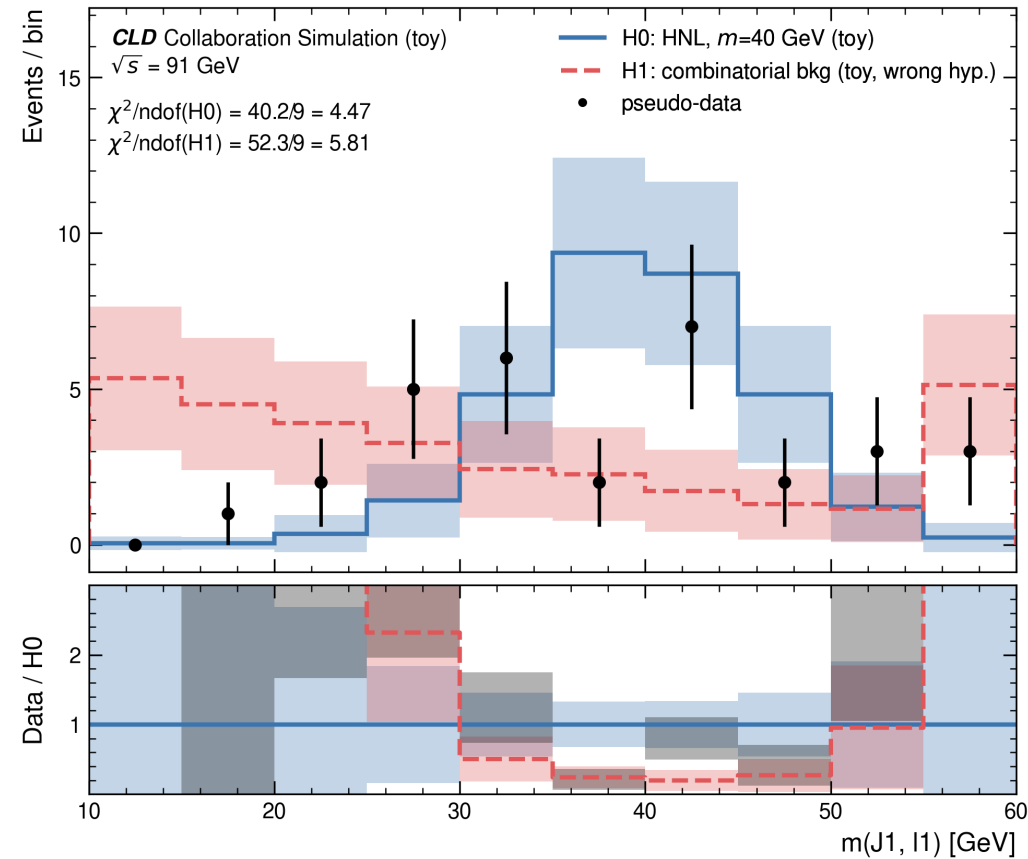
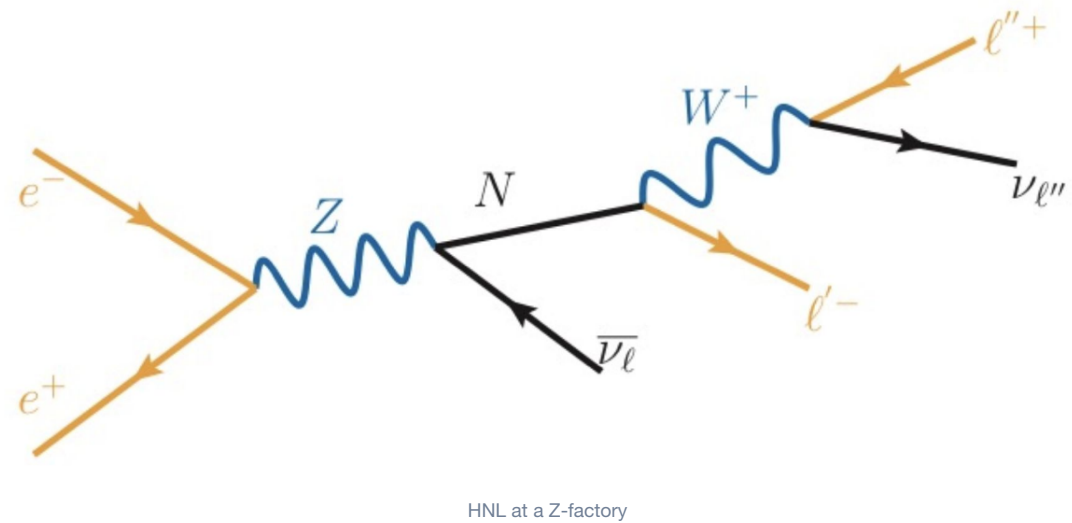
Next: **Andrea Maria**

Sum every identified SM process and compare to the reported points. At both **91** and **365 GeV** the excess sits far above that sum. This is why an interpretation is needed — step 2 becomes step 3.

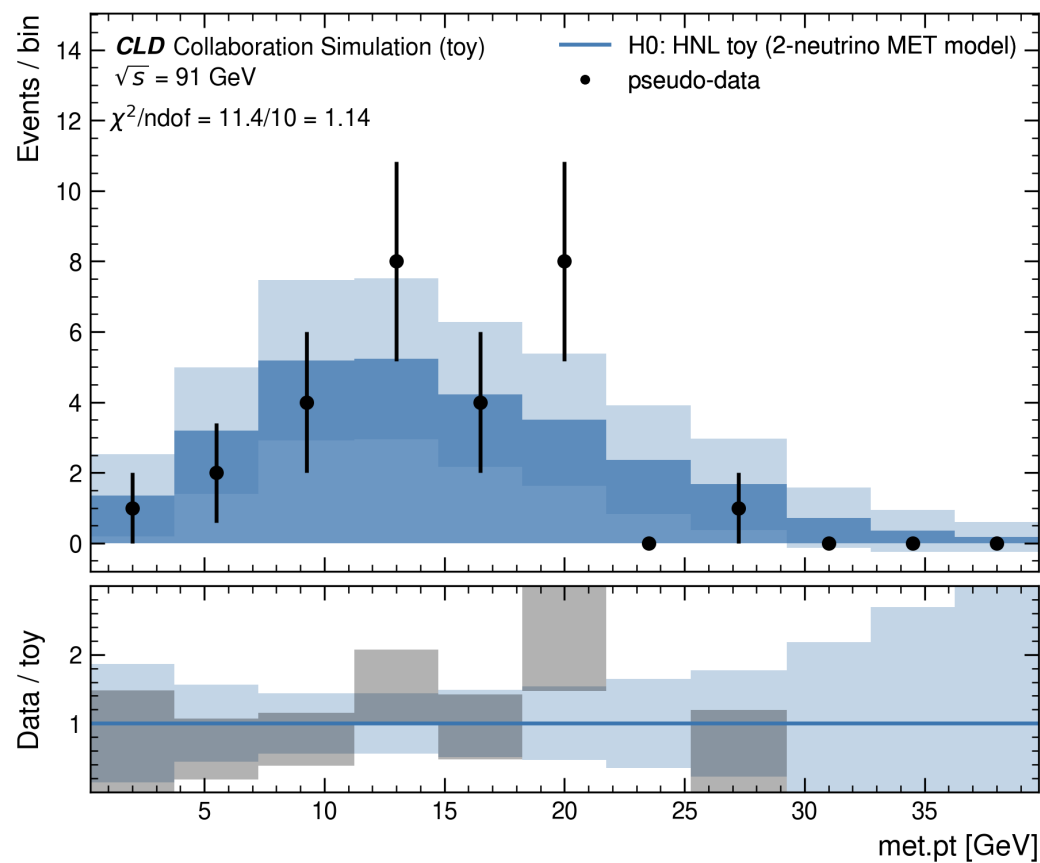


Excess → Reconstruction → H1: flat bkg → H0: HNL, $m \approx 40$ → Test

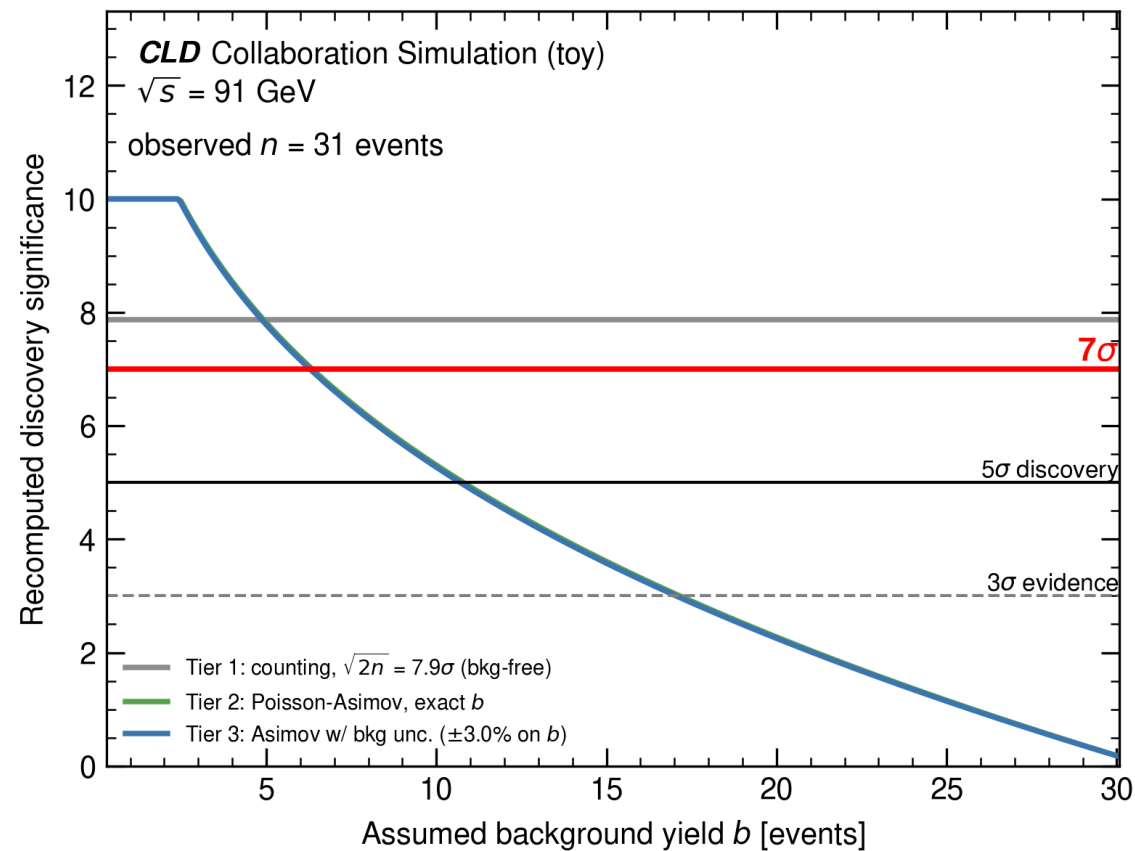
Tight selection: MET $p_T > 3$ GeV, ≥ 1 lepton, small impact parameter. A lepton+jet excess appears. Proposed: a **Heavy Neutral Lepton**, mass ≈ 40 GeV. The resonance hypothesis fits better than a flat background: χ^2/ndof 4.47 vs. 5.81.



A second, independent variable — missing energy itself — agrees with the same hypothesis: $\chi^2/\text{ndof} = 1.14$. This isn't the same test twice.



The source quotes $\sim 7\sigma$ as its headline number. Our background-free recomputation gives 7.9σ — close, but we can't confirm they're the same definition. The source also quotes 5.79σ and 7.8σ elsewhere; we don't know which tier those are (see audit). No look-elsewhere correction is applied: we don't have a real trials count.

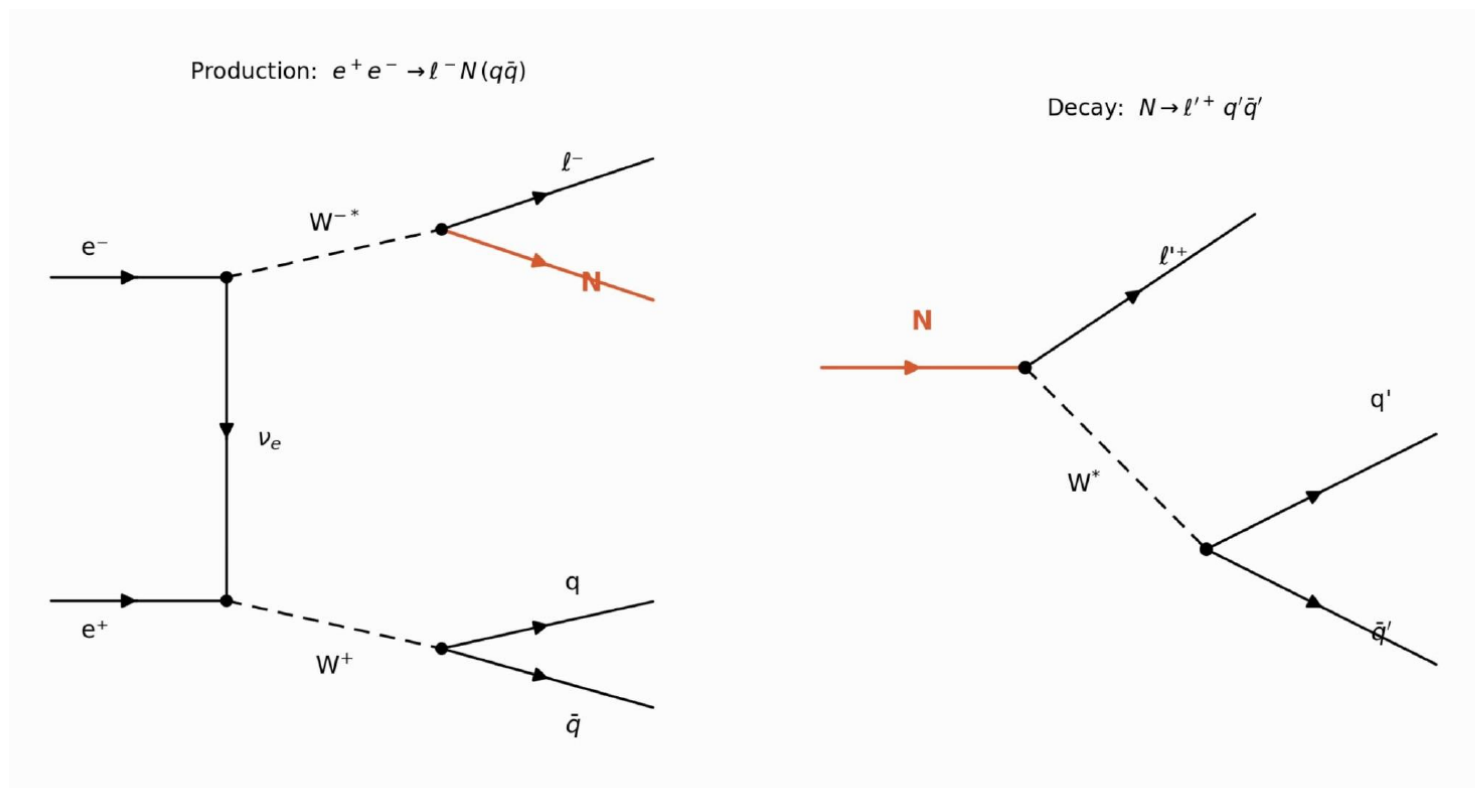


The excess at 365 GeV

Testing reconstruction and background hypotheses

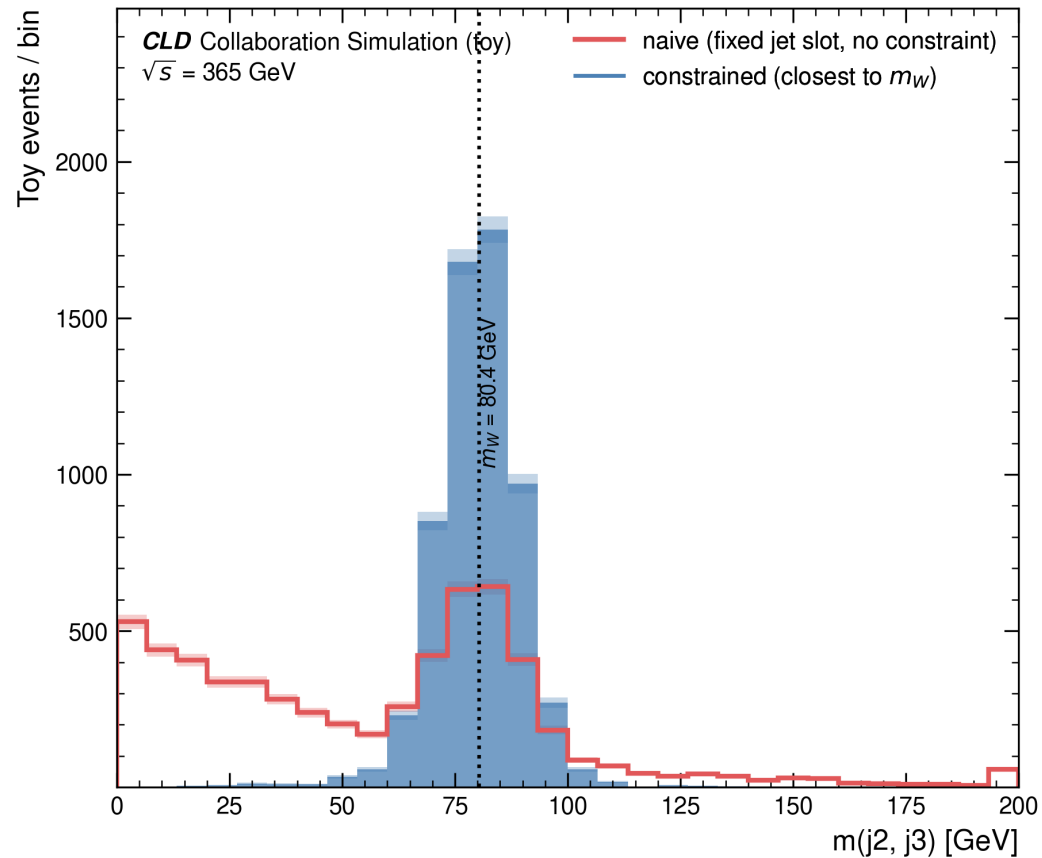
Next: **Jurjan Bootsma**

Tighter selection again: low MET, **≥ 2 jets**, **≥ 2 leptons**, Z-veto, b-veto. Original method: hadronic-W channel, one jet assumed, a fixed pairing $m(j_2, j_3)$. The source itself flags this: “*sub-optimal pairing might lead to large spread.*”

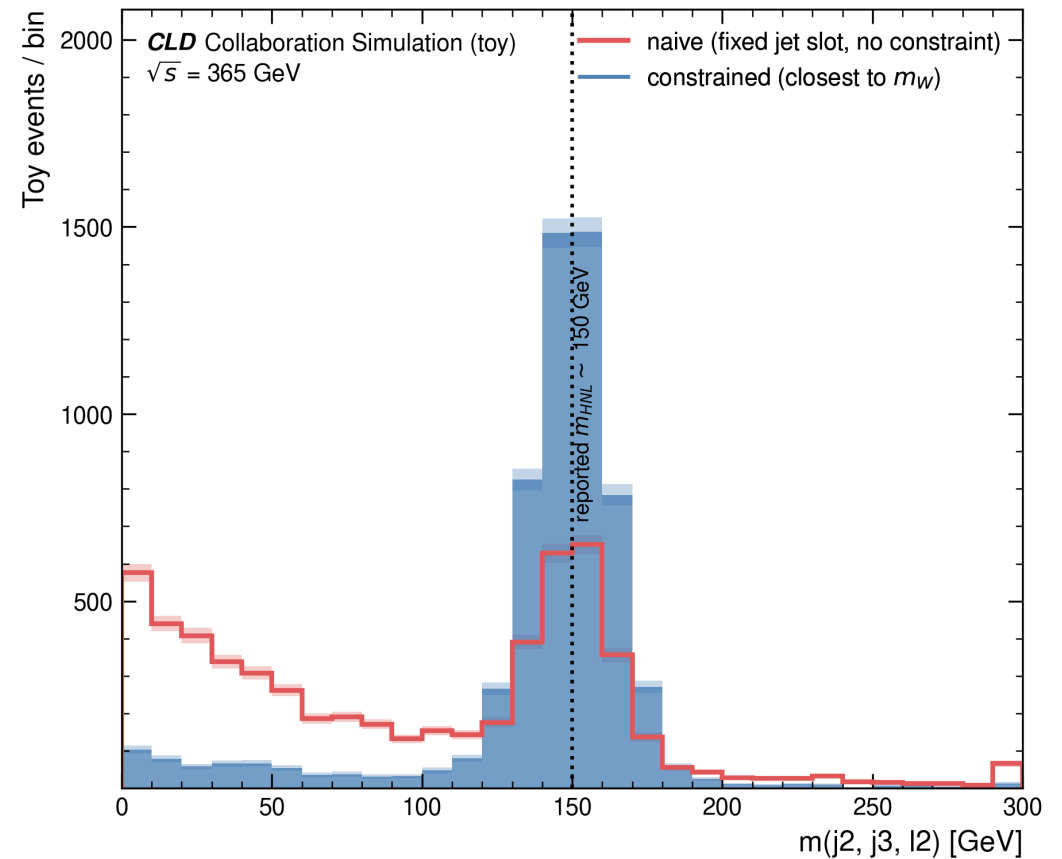


Production and decay, slide 18

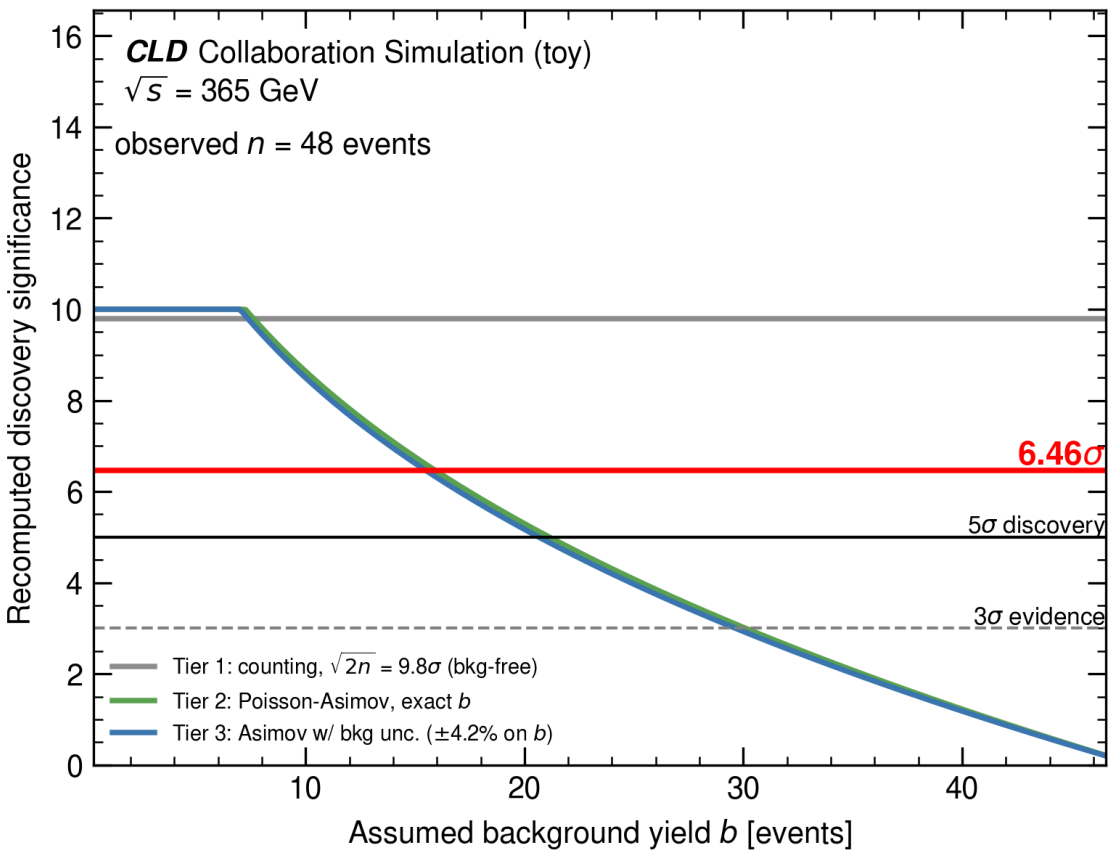
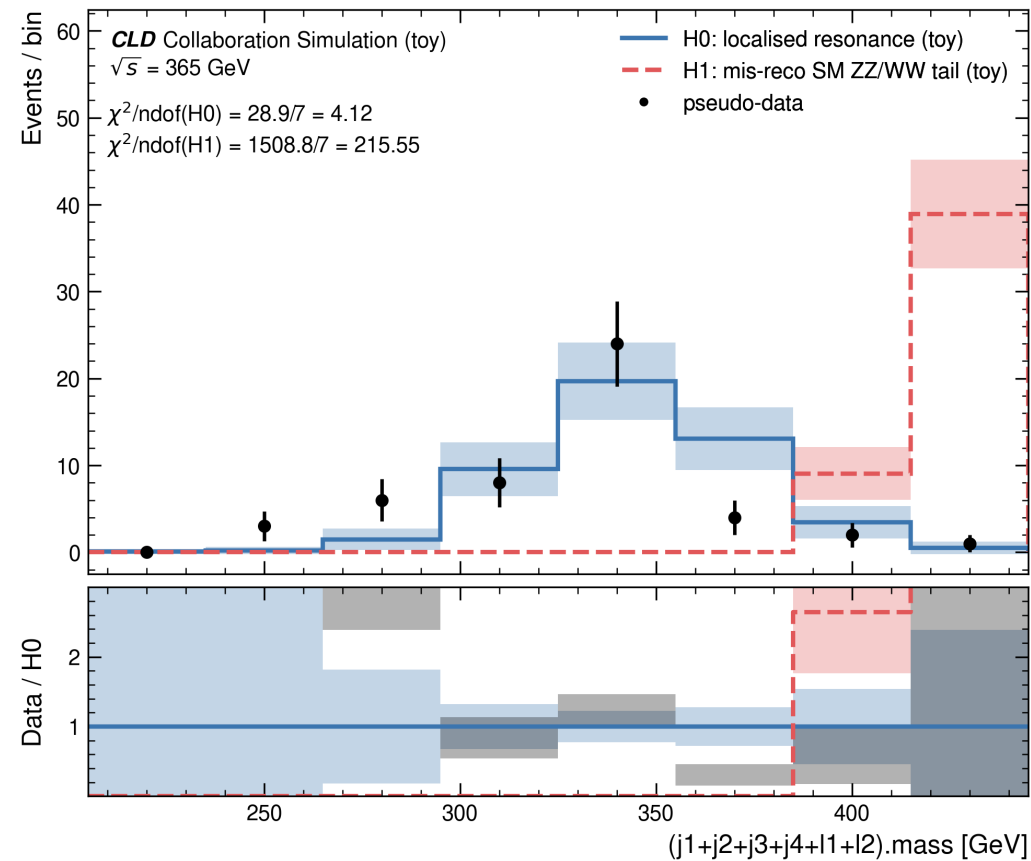
Four jets give three candidate pairings. The fixed original pairing is correct only **33%** of the time by construction. Picking the pairing closest to m_W instead gets it right **88%** of the time, and both peaks narrow sharply. This is evidence the original pairing can mis-associate jets — it supports better reconstruction, not an HNL by itself.



Naive = original · Constrained = current analysis



Resonance vs. a mis-measured ZZ/WW tail: χ^2/ndof **4.12 vs. 215.55** — this specific SM-tail model is disfavoured. Quoted significance: **6.46 σ** . The background behind that number isn't established in the source.



What has been established -- and what remains open?

Consistency, caveats, and conclusions

Next: **Greta Brianti**

- **40 GeV vs. 150 GeV**: two HNL masses. One particle has one mass. This is open, not explained away.
- 365 GeV background normalisation: not established.
- No look-elsewhere trial count is available, so none is applied.
- Every cut efficiency shown is a toy assumption.
- Rejecting one background model is not proof of an HNL.

All five 365 GeV processes and both 160 GeV processes now have a toy-vs-pseudo-data comparison — every value traceable to this repo's code.

Process (365 GeV)	χ^2/ndof
$\bar{t}t$	0.81
$ZZ \rightarrow \ell\ell q\bar{q}$	8.40
$e^+e^- \rightarrow \bar{f}f$	5.17†
ZH	3.89†
WW	26.22†
Process (160 GeV)	χ^2/ndof
Higgs	37.77
WW	1.84

† second-pass digitisation, lower precision than the other rows (see PROVENANCE_AUDIT.md).

At both energies the resonance hypothesis fits better than the flat/SM-tail alternative tested. That supports the interpretation. It doesn't prove it.

	91 GeV	365 GeV
Resonance (H0) χ^2/ndof	4.47	4.12
Flat / SM-tail (H1) χ^2/ndof	5.81	215.55
Reconstructed mass	≈ 40 GeV	≈ 150 GeV
Quoted significance	5.79 / 7 / 7.8 σ	6.46 σ
Jet mis-association (365 only)	—	33% $\rightarrow$ 88%

Five SM processes at 365 GeV, two at 160 GeV, each traceable. Two excesses, both **consistent with** a resonance hypothesis, both with open caveats. FCC-ee's precision lets small, careful excesses be taken seriously — if every claim stays traceable. So: **what do you think is really in there?**

Which process do you think could be among the BSM processes studied?

A Heavy Neutral Lepton (HNL)

B Leptoquark

C Dark photon / Z'

D Supersymmetric slepton

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